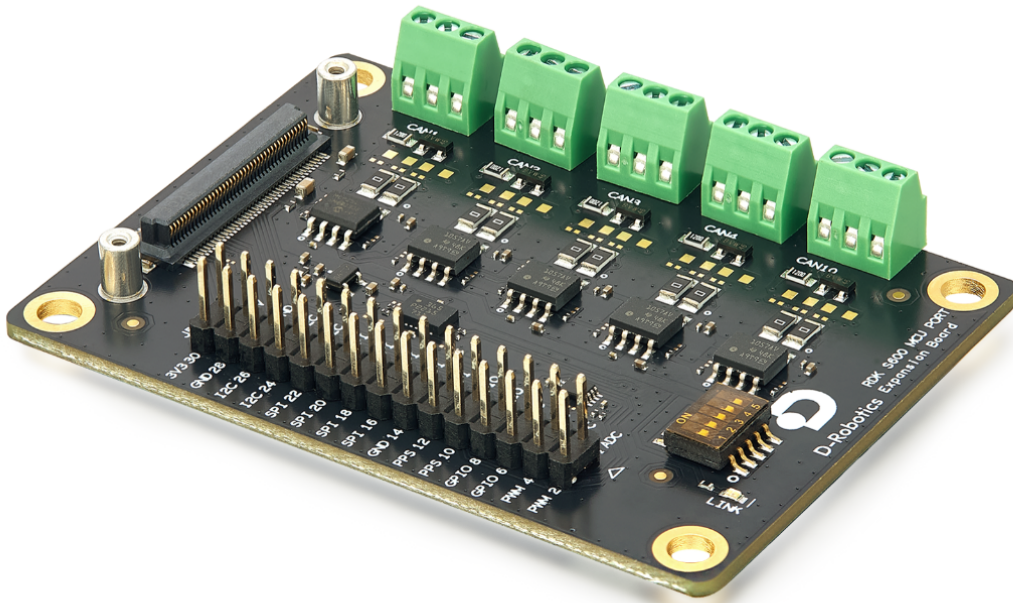


MCU Interface Expansion Board

Product Documentation URL: https://developer.d-robotics.cc/rdk_s_doc/en/Quick_start/hardware_introduction/rdk_s600/rdk_s600_mcu_port_expansion_board?v=5.1.0&p=RDK+S600

1.Product Introduction



The RDK S600 MCU Port Expansion Board (including the matching FPC) is a core expansion module for the D-Robotics RDK S600 series developer kit. It is mainly used to expand MCU interface functions, supporting CAN_FD, ADC, etc.

warning

- This product is only compatible with the RDK S600 series mainboard. Do not use it with other device models.
- During use, place the expansion board on a stable, flat, and non-conductive surface to avoid the device falling or short-circuiting due to unstable support.
- If an incompatible device is connected to this MCU expansion board, this product does not provide repair services for any resulting device damage.
- All peripheral devices used with this product (including but not limited to CAN devices) must comply with the safety and performance standards of the country/region of use and bear appropriate compliance certification marks.
- All cables and connectors of peripheral devices connected to this expansion board must have sufficient insulation performance to meet electrical safety requirements.

Safe Usage

To avoid malfunction or damage to this expansion board, strictly follow these guidelines:

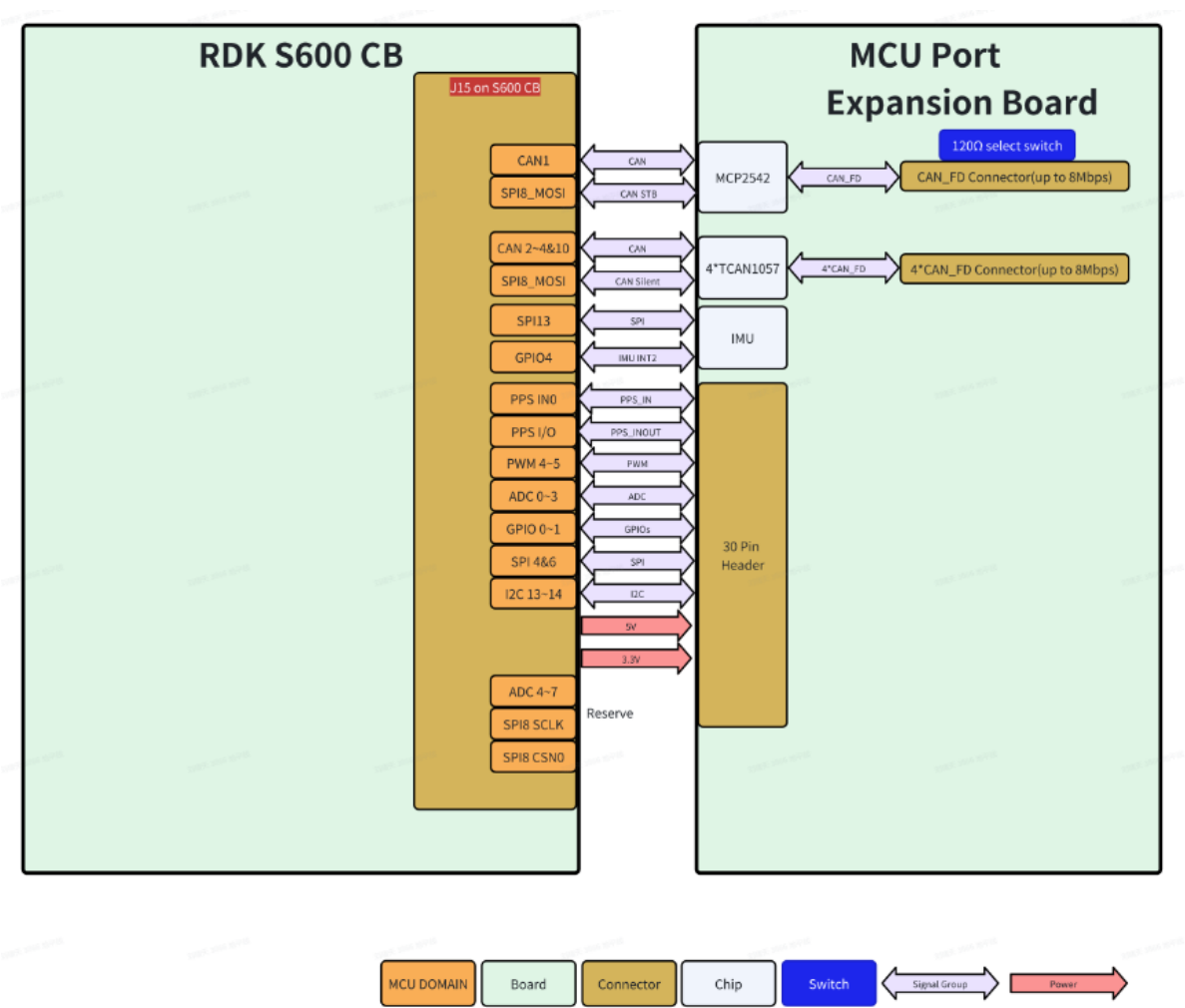
- **Environmental Requirements:** Do not expose to water, moisture, or conductive surfaces during operation. Keep away from heat sources (e.g., heaters, direct sunlight). Ensure the operating environment temperature complies with the product specification requirements.

- **Assembly Operations:** Handle with care during assembly. Avoid applying mechanical pressure or electrical interference (such as electrostatic discharge) to the printed circuit board (PCB) and connectors.
- **Power-On Operations:** Do not touch the PCB surface or metal interfaces on the edge of the device while powered on to reduce the risk of electrostatic discharge (ESD) damage.

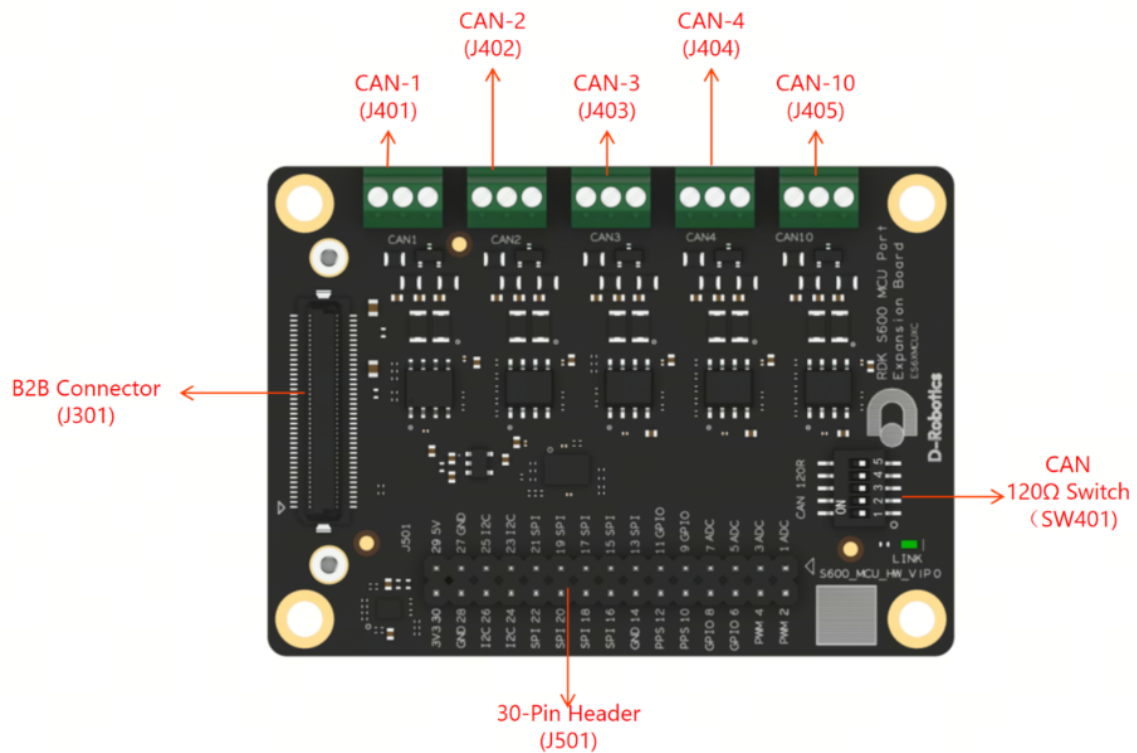
2.Product Specifications

Item	Parameters
Interfaces	5 x CAN FD (up to 8Mbps) 1 x 30-pin, supporting up to 7x ADC, 2x IIC, 2x SPI
Onboard Module	IMU: BMI088 (SPI)
Operating Temp	0°C~65°C
Dimensions	70x70x17mm

Architecture Diagram



Interface Description



Reference	Function
J301	MCU Expansion Board 80-pin Connector
J401	MCU-CAN1 Interface
J402	MCU-CAN2 Interface
J403	MCU-CAN3 Interface
J404	MCU-CAN4 Interface
J405	MCU-CAN10 Interface
J501	30-Pin Interface
SW401	MCU CAN 120Ω Switch

Assembly Instructions

Important Reminders

- Perform the installation only when the development board is powered off and the DC plug is disconnected.
- Keep the connectors aligned parallel during assembly. Mate the connectors evenly with consistent force and ensure a secure connection to prevent damage.

Tips

The side of the FPC cable marked with "CB" on the silkscreen corresponds to the J15 connector on the RDK S600 mainboard. The side marked with "SUB" corresponds to the J301 connector on this expansion board.

3.Interface Details

CAN FD Connectors (J401/J402/J403/J404/J405)

The back of the expansion board indicates `CAN_H`, `CAN_L`, and `GND` for each interface.

The expansion board provides 5 CAN FD interfaces (CAN-1~CAN-4, CAN-10). Each interface is equipped with a 120Ω terminal resistor, which can be toggled via the switch (SW401).

30-Pin (J501)

Pin definition: [drobotics_rdk_s600_mcu_port_expansion_board_pinlist_v1p0.xlsx](#)

- When using the following 5 IOs from the 30-Pin Connector with peripherals, ensure that the default power-on high/low state of the corresponding peripheral pins matches the Pull Up/Down state specified in the Pin Definition file. Do not add or connect additional pull-up/pull-down resistors:
 - PIN11: MCU_GPIO0_3V3
 - PIN15: MCU_SPI4_CSN0_3V3
 - PIN19: MCU_SPI4_MOSI_3V3
 - PIN16: MCU_SPI6_CSN0_3V3
 - PIN20: MCU_SPI6_MOSI_3V3
 - PIN13: MCU_SPI4_CSN1_3V3

IMU (U301)

Integrated Inertial Measurement Unit (IMU, model Bosch Sensortec BMI088), supporting communication control via the SPI-13 serial bus.

4.Indicator Light

Below the 5 switches (SW401) on the expansion board, there is one green LED indicator (labeled "LINK") used to indicate the power status and the connection status between the MCU expansion board and the RDK S600:

- Green light constantly on: The RDK S600 and MCU expansion board are connected normally, and 5V power is being supplied correctly.
- Green light off: The connection between the RDK S600 and MCU expansion board is abnormal, with no 5V power supply.